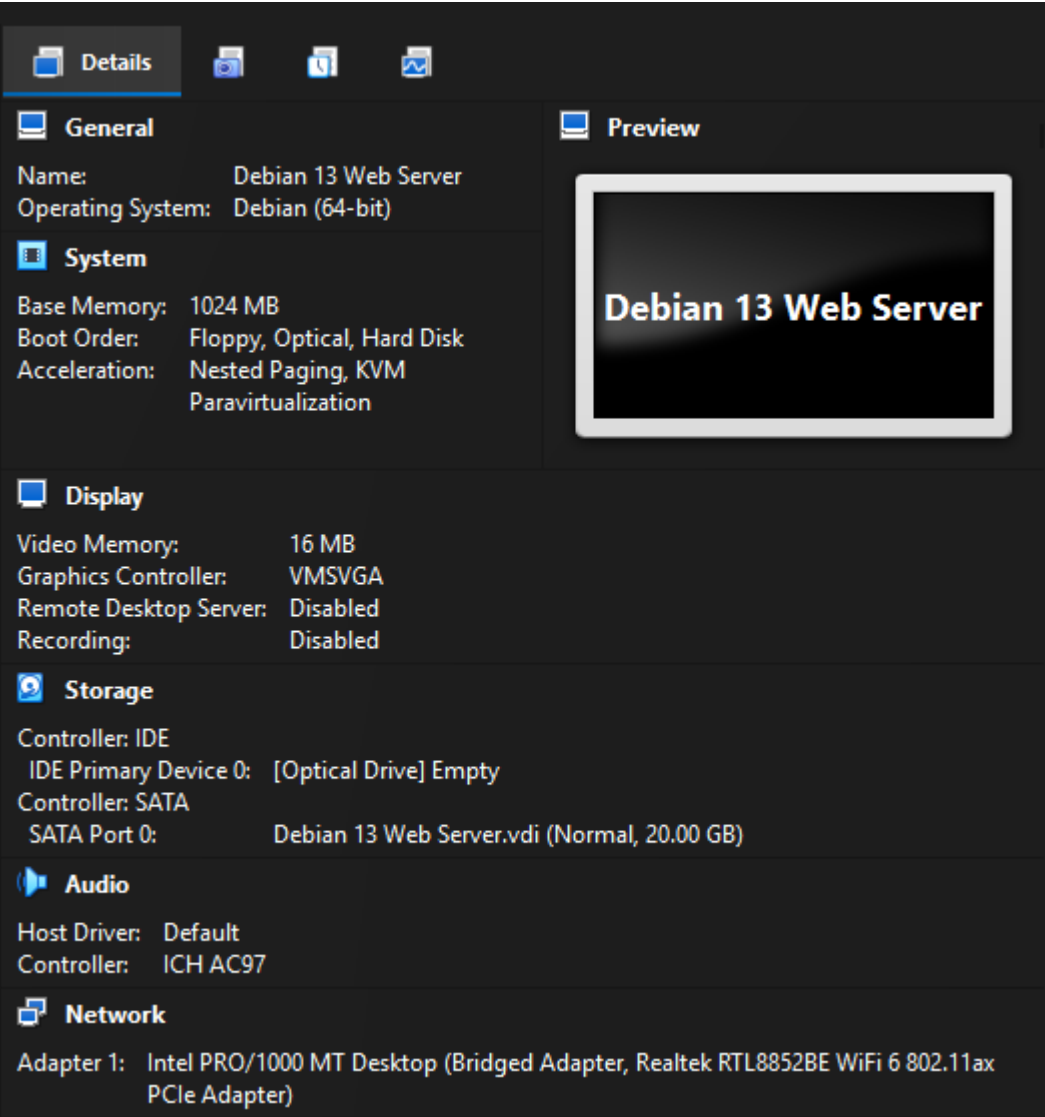


Deliverable 2

Questions

1. What are the server hardware specifications (virtual machine settings)?



1. What is the Debian Login Screen?

```
Debian GNU/Linux 13 debianws tty1
debianws login: lenjrg13
Password:
Linux debianws 6.12.57+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.57-1 (2025-11-05) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
```

2. What is the IP address of your Debian Server Virtual Machine? Here is a screenshot:

```
lenjrg13@debianws:~$ hostname -I  
192.168.1.59 2600:4809:98d6:1400
```

3. How do you work with the Firewall in Debian? A network firewall is a security system, hardware or software, that acts as a digital barrier, monitoring and controlling incoming and outgoing network traffic to protect a private network from unauthorized access and cyber threats like hackers, viruses, and malware by enforcing predefined security rules.

- How do you check if the Firewall is running? Open a terminal and run the following command: `sudo ufw status`
- How do you disable the Firewall? Open a terminal and run the following command: `sudo ufw disable`
- How do you add Apache to the Firewall? Open a terminal and run the following command to identify which profiles are available: `sudo ufw app list | grep Apache` Then, run `sudo ufw allow "Profile Name"`.

1. What different commands do we use to work with Apache?

- What is the command you use to check if Apache is running?
 - To check if Apache is running, open a terminal and run the following command: `systemctl status apache.service` as shown in the following screenshot.

```
lenjrg13@debianws:~$ systemctl status apache2.service  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: ▶  
   Active: active (running) since Wed 2025-12-10 08:23:33 EST; 3min 2s ago  
 Invocation: c4bc9d022b2842638ea68a28fa6141f1  
    Docs: https://httpd.apache.org/docs/2.4/  
  Process: 742 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUC▶  
 Main PID: 809 (apache2)  
    Tasks: 55 (limit: 1110)  
  Memory: 7.8M (peak: 8.1M)  
     CPU: 160ms  
   CGroup: /system.slice/apache2.service  
           └─809 /usr/sbin/apache2 -k start  
             └─810 /usr/sbin/apache2 -k start  
               └─812 /usr/sbin/apache2 -k start
```

- What is the command you use to stop Apache?
 - To stop Apache from running, open a terminal and run the following command: `systemctl status apache.service` as shown in the following screenshot.

```

root@debianws:~# systemctl stop apache2.service
root@debianws:~# systemctl status apache2.service
o apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: >
   Active: inactive (dead) since Wed 2025-12-10 08:33:22 EST; 19s ago
   Duration: 9min 48.968s
   Invocation: c4bc9d022b2842638ea68a28fa6141f1
   Docs: https://httpd.apache.org/docs/2.4/
   Process: 742 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUC>
   Process: 3442 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, stat>
   Main PID: 809 (code=exited, status=0/SUCCESS)
   Mem peak: 9.6M
   CPU: 437ms

Dec 10 08:23:33 debianws systemd[1]: Starting apache2.service - The Apache HTTP>
Dec 10 08:23:33 debianws apachectl[806]: AH00557: apache2: apr_sockaddr_info_ge>
Dec 10 08:23:33 debianws apachectl[806]: AH00558: apache2: Could not reliably d>
Dec 10 08:23:33 debianws systemd[1]: Started apache2.service - The Apache HTTP >
Dec 10 08:33:22 debianws systemd[1]: Stopping apache2.service - The Apache HTTP>
Dec 10 08:33:22 debianws apachectl[3444]: AH00558: apache2: Could not reliably >
Dec 10 08:33:22 debianws systemd[1]: apache2.service: Deactivated successfully.
Dec 10 08:33:22 debianws systemd[1]: Stopped apache2.service - The Apache HTTP >

```

- What is the command you use to restart Apache?
 - To restart a running service, you must open a terminal and run the following command:
`systemctl restart apache2.service` as shown in the screenshot below.

```

root@debianws:~# systemctl reload apache2.service
root@debianws:~# systemctl status apache2.service
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: >
   Active: active (running) since Wed 2025-12-10 08:35:57 EST; 7s ago
   Invocation: 1336e5ec68334fc5b848326545c3643f
   Docs: https://httpd.apache.org/docs/2.4/
   Process: 4115 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SU>
   Process: 4192 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=>
   Main PID: 4118 (apache2)
   Tasks: 55 (limit: 1110)
   Memory: 4.8M (peak: 7M)
   CPU: 208ms
   CGroup: /system.slice/apache2.service
           └─4118 /usr/sbin/apache2 -k start
             └─4197 /usr/sbin/apache2 -k start
               └─4198 /usr/sbin/apache2 -k start

```

- What is the command used to test Apache configuration?
- Use the following command `sudo apache2ctl configtest` as shown in the following screenshot.

```

root@debianws:~# apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name
, using 127.0.1.1. Set the 'ServerName' directive globally to suppress this message
Syntax OK

```

- What is the command used to check the installed version of Apache?
 - Use the following command to check the installed version of an application: `apache2ctl -v`

```

root@debianws:~# apache2ctl -v
Server version: Apache/2.4.65 (Debian)
Server built: 2025-07-29T17:52:31

```

- What are some common configuration files for Apache?

- I was able to identify the following files: {access, error, other_vhosts_access}.log files after running `ls /var/log/apache2` in the terminal.

```
root@debianws:~# ls -l /var/log/apache2/
total 8
-rw-r----- 1 root adm 170 Dec  9 12:06 access.log
-rw-r----- 1 root adm 2449 Dec 10 08:36 error.log
-rw-r----- 1 root adm  0 Dec  9 11:54 other_vhosts_access.log
```

- Where does Apache store logs?
 - Apache stores log in the following directory `/var/log/apache2`
- What are some basic commands we can use to review logs?
 - The `cat` command can be used to review log files. For example: `cat /var/log/apache2/access.log`

```
root@debianws:~# cat /var/log/apache2/
access.log          error.log          other_vhosts_access.log
root@debianws:~# cat /var/log/apache2/error.log
[Tue Dec 09 11:56:49.557253 2025] [mpm_event:notice] [pid 693:tid 693] AH00489: Apache/
2.4.65 (Debian) configured -- resuming normal operations
[Tue Dec 09 11:56:49.557355 2025] [core:notice] [pid 693:tid 693] AH00094: Command line
: '/usr/sbin/apache2'
[Tue Dec 09 11:57:42.204541 2025] [mpm_event:notice] [pid 694:tid 694] AH00489: Apache/
2.4.65 (Debian) configured -- resuming normal operations
[Tue Dec 09 11:57:42.215049 2025] [core:notice] [pid 694:tid 694] AH00094: Command line
: '/usr/sbin/apache2'
[Tue Dec 09 11:58:52.909826 2025] [mpm_event:notice] [pid 695:tid 695] AH00489: Apache/
2.4.65 (Debian) configured -- resuming normal operations
[Tue Dec 09 11:58:52.913850 2025] [core:notice] [pid 695:tid 695] AH00094: Command line
: '/usr/sbin/apache2'
[Tue Dec 09 12:01:01.126337 2025] [mpm_event:notice] [pid 696:tid 696] AH00489: Apache/
2.4.65 (Debian) configured -- resuming normal operations
[Tue Dec 09 12:01:01.133599 2025] [core:notice] [pid 696:tid 696] AH00094: Command line
: '/usr/sbin/apache2'
```